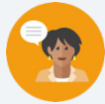


4.7 Justifying Congruence

Lesson Introduction

 Benchmarks	MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure. MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.		 Learning Targets	- I can describe a sequence of transformations that will map a preimage and an image to show congruence. - I can describe a sequence of transformations that map one figure onto another congruent figure.
 Benchmark Coherence	Building On MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.		Working Toward N/A	
 Essential Questions	- How can you describe a sequence of transformations that will map a preimage and an image? - How can a sequence of transformations be used to show congruence?			
 Lesson Narrative	Lesson 7 builds on a working definition of congruence to include rigid motion. Students justify that two shapes are congruent (preimage and image) by applying the definition of congruence in terms of rigid motions. Each activity incorporates elements of flexibility, as students can choose which shape is the preimage, providing pathways for a variety of responses.			
 Component Summary	4.7.1 Warm-Up (~5 min.)	Goal-setting activity (<i>SE p. 213</i>)	4.7.3 Guided Practice (10-15 min.)	Practice justifying congruence (<i>SE p. 215</i>)
	4.7.2 Guided Instruction (15-20 min.)	Justifying congruence through rigid motions (<i>SE p. 213</i>)	4.7.4 Your Turn! (5 min.)	Describe a sequence of transformations using algebraic descriptions (<i>SE p. 216</i>)
	4.7.5 Wrap-Up (~5 min.)	Ticket Out the Door describing a sequence of transformations to justify congruence (<i>SE p. 217</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> Definition of Congruence in Terms of Rigid Motions <u>Additional Terms:</u> N/A		- Growth Mindset video (Warm-Up) (Optional) Patty paper	4.7.1 Warm-Up contains a video to accompany the question prompt.

 Teacher Tips	Common Misconceptions - Students may recognize the correct transformations but do them in an order that results in the preimage not mapping onto the image. - Students may incorrectly write the transformation algebraically. - Students may not match the corresponding vertices correctly.	Things to Consider The 4.7.1 Warm-Up provides an opportunity to incorporate practices of growth mindset into your lesson. Integrate methods like positive talk, feedback loops, or sentence stems for students to use during each of the activities in the lesson.
	Literacy and Language Routines Stronger and Clearer Each Time During 4.7.2 Guided Instruction, students will justify congruence through rigid motion. Provide students multiple opportunities to articulate their responses, gathering feedback from peers while incorporating that feedback for more refined and precise mathematical answers.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.3.1: Mathematical Fluency Students have the opportunity to identify and share their uniquely different yet correct sequences of transformations in each activity. Because preimage and image are not identified, students' approaches will vary, providing opportunity to position students as experts of their own work, building both confidence and fluency.
	 Differentiate Instruction	
The 4.7.1 Warm-Up provides an opportunity to incorporate practices of growth mindset into your lesson. Integrate methods like positive talk, feedback loops, or sentence stems for students to use during each of the activities in the lesson. - Ask students what James Clear's quote means to them or could mean in a particular situation in their academic life currently. - Create accountability partners for students to identify moments of growth from current results and have students revisit these goals and end results in Unit 5. - Provide sentence stems for students to incorporate positive self-talk into each activity, pausing after each activity to give students 60 seconds to turn and reflect with a partner using positive self-talk.		
Lesson Notes:		

4.7.1 Warm-Up: You Can Do It!

Component Overview: Students will complete an activity focused on setting goals. Consider showing the video prior to having students complete their goals.



4.7.1 Warm-Up: You Can Do It!

"You should be far more concerned with your current trajectory than with your current results." - James Clear

Has anyone ever asked you, "What goals do YOU want to achieve in your lifetime?"



Setting goals is linked to self-confidence, motivation, and autonomy. A 2015 study showed people who wrote down their goals were 33 percent more successful in achieving them than those who just thought about the outcomes. Making goals can be fun.

For example, you can name places you would like to visit, activities you would like to do, or people you would like to meet. By setting the goal, you have already set your trajectory toward achieving it. In the table, write three life goals and 2 goals for the rest of this school year.

Life Goals	Goals for this school year
<i>Answers will vary.</i>	<i>Answers will vary.</i>
"If you are always trying to be normal, you will never know how amazing you can be." Maya Angelou	



Consider building on this activity by having students set goals for this lesson particular lesson, this unit, or this class.

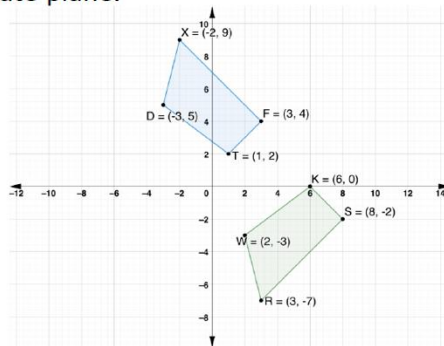
4.7.2 Guided Instruction: Justifying Congruence

Component Overview: Students will describe sequences of transformations that map quadrilaterals onto one another. They will justify congruence of the two quadrilaterals by describing the mapping so that the definition of congruence in terms of rigid motions can be applied. For this activity, employ a gradual release model to guide students through the Guided Instruction.



4.7.2 Guided Instruction: Justifying Congruence

1. Quadrilaterals $XFTD$ and $KSRW$ are shown on the coordinate plane.



- a. Describe a sequence of transformations that will map quadrilateral $KSRW$ onto quadrilateral $XFTD$.
*Answers will vary. A vertical translation down 2 units, a horizontal translation left 5 units, then a reflection across the x -axis.
 A reflection across the line $y = 1$ with a horizontal translation left 5 units.
 A vertical translation up 2 units, horizontal translation left 5 units, then a reflection across the line $y = 2$.*
- b. Complete the table that shows which vertices were mapped to each other when quadrilateral $KSRW$ was mapped onto quadrilateral $XFTD$.

Quadrilateral $KSRW$	Quadrilateral $XFTD$
K	T
S	F
R	X
W	D



Consider an alternate entry point of having students create a table of values for each quadrilateral. Ask the students to informally describe the transformations that they think occurred. Then focus on one vertex and apply the transformations using the ordered pairs or map them on the grid.



How do you determine which vertex from one quadrilateral corresponds to which vertex from the other? Why would the congruence statement be wrong if incorrect vertices were mapped?

4.7.2 Guided Instruction: Justifying Congruence (continued)



Definition of Congruence in Terms of Rigid Motions

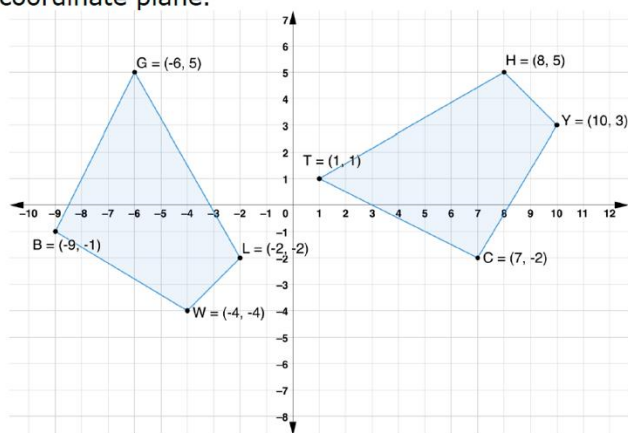
Two figures are congruent if and only if there exists one or more rigid motions, which will map one figure onto the other.

- c. Complete the statements.

Quadrilateral $KSRW$ is congruent to quadrilateral $XFTD$ because the sequence of transformations that can be used to map the quadrilaterals were **rigid motions**.

Because vertex K maps to vertex T , vertex S maps to vertex F , vertex R maps to vertex X and vertex W maps to vertex D . The definition of congruence in terms of rigid motions justifies that $KSRW \cong TFXD$.

2. Quadrilaterals $BGLW$ and $HYCT$ are shown on the coordinate plane.



Stronger and Clearer

The formal definition of congruence in terms of rigid motions was introduced in the prior lesson. Consider giving students an opportunity to make sense of information using the example and prompt. Have students incorporate an “if and only if” meaning within their responses.



For question 1c, refer to the definition to point out the conditions of the definition of congruence in terms of rigid motions were met. Avoid stating that congruence exists because the figures are the same shape and the same size.



Consider an inquiry-based approach to Guided Instruction. What transformations do you think occurred with quadrilaterals $BGLW$ and $HYCT$? Which quadrilateral did you use as the preimage? How could choosing the other quadrilateral as the preimage change the transformations?

4.7.2 Guided Instruction: Justifying Congruence (continued)

- a. Use algebraic descriptions to describe a sequence of transformations that will map quadrilateral $BGLW$ onto quadrilateral $HYCT$.

Answers will vary.

$$(x, y) \rightarrow (-y, x)$$

$$(x, y) \rightarrow (x + 6, y + 7)$$

OR

$$(x, y) \rightarrow (-y, -x)$$

$$(x, y) \rightarrow (x, -y)$$

$$(x, y) \rightarrow (x + 6, y + 7)$$

- b. Complete the table that shows which vertices were mapped to each other when quadrilateral $BGLW$ was mapped onto quadrilateral $HYCT$.

Quadrilateral $BGLW$	Quadrilateral $HYCT$
B	C
G	T
L	H
W	Y



When using rigid motions to map one figure onto the other, always be sure the corresponding vertices match before stating congruence.

- c. Complete the congruence statement.

quadrilateral $BGLW \cong$ *quadrilateral* $CTHY$



MA.K12.MTR.3.1: Mathematical Fluency

Notice the flexible pathways for responses since the question does not specify preimage and image. Provide opportunity for students to build flexibility in their approaches by monitoring for and sharing student responses that are different.



- Why would being sure that the corresponding vertices match before stating congruence be an important step in this process?
- Does the order the vertices are listed matter when writing a congruence statement?

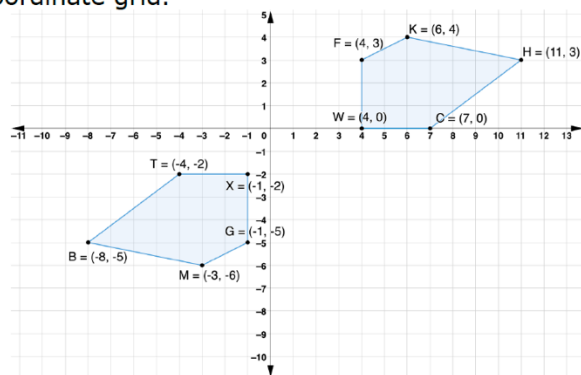
4.7.3 Guided Practice: Justifying Congruence

Component Overview: Students will engage in sequences of transformations to a preimage onto an image. They will practice using this to justify congruence of the two figures. For this activity, release students to work with a partner to practice these questions.



4.7.3 Guided Practice: Justifying Congruence

1. Polygons $TXGMB$ and $WFKHC$ are shown on the coordinate grid.



- a. Use algebraic descriptions to describe a sequence of transformations that will map polygon $TXGMB$ onto polygon $WFKHC$.

Answers will vary.

$$(x, y) \rightarrow (-x, -y)$$

$$(x, y) \rightarrow (x + 3, y - 2)$$

OR

$$(x, y) \rightarrow (-x, y)$$

$$(x, y) \rightarrow (x, -y)$$

$$(x, y) \rightarrow (x + 3, y - 2)$$

- b. Complete the congruence statement.

polygon $TXGMB \cong$ *polygon CWFKH*



Consider doing a *Think Aloud* to launch this activity. This can help students better understand what to look for when preparing to complete the activity. Allow students to share the transformations they see, making connections between visual and verbal descriptions and their algebraic representations.



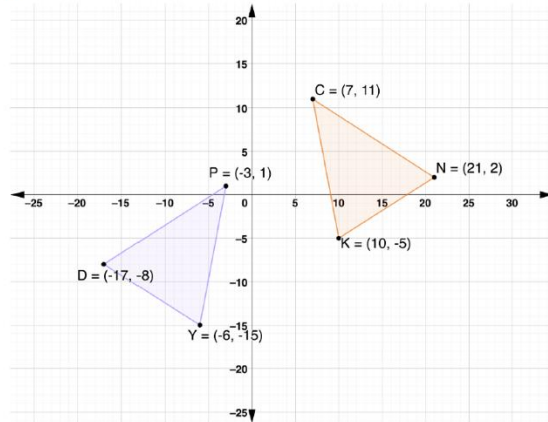
Ensure students understand that there can be multiple sequences of transformations that show congruence.



Monitor for and have students share work that highlights different methods (tables of values, mapping, written descriptions) that can be used to write the algebraic descriptions.

4.7.3 Guided Practice: Justifying Congruence (continued)

2. Triangles CKN and PYD are shown on the coordinate grid.



- a. The sequence of transformations that will map $\triangle CKN$ onto $\triangle PYD$ is

- ☐ translating $\triangle CKN$ left 14,
- ☐ reflecting $\triangle CKN$ across the y -axis,
- ☐ rotating $\triangle CKN$ 270° clockwise about the origin,

then

- ☐ translating $\triangle C'K'N'$ right 4, down 10.
- ☐ translating $\triangle C'K'N'$ right 10, down 4.
- ☐ reflecting $\triangle C'K'N'$ across the x -axis.

- b. Vertex C maps to vertex P , vertex K maps to vertex Y , and vertex N maps to vertex D therefore $\triangle CKN$ is congruent to $\triangle PYD$ by the definition of congruence in terms of rigid motions.



In question 2a, were you able to quickly eliminate any of the options in the first box? In the second box? Explain your rationale.



For a challenge, have students attempt to find a way to make one of the incorrect answers correct using information not provided as one of the initial answer choices.

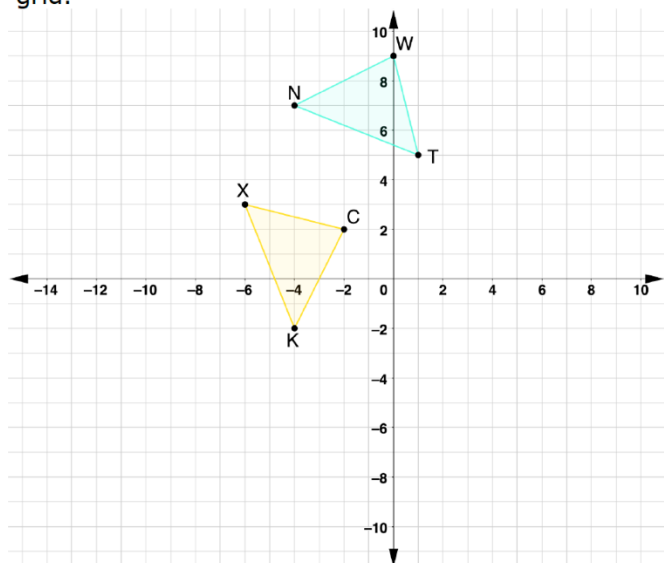
4.7.4 Your Turn!

Component Overview: Students will work independently to write algebraic descriptions to describe a sequence of three transformations that will map a triangle onto another.



4.7.4 Your Turn!

1. Triangles XCK and TWN are shown on the coordinate grid.



Use algebraic descriptions to describe a sequence of three transformations that will map $\triangle XCK$ onto $\triangle TWN$.

Transformations
$(x, y) \rightarrow (-x, y)$
$(x, y) \rightarrow (y, -x)$
$(x, y) \rightarrow (x - 2, y + 11)$



Consider including a “my favorite no” to highlight creative thinking that can highlight common misconceptions and provide opportunity for clarification.



Consider allowing students to answer with a written/verbal description of the transformations to check their understanding of the rigid motions. Then use the tables of the algebraic descriptions from previous lessons to provide support in making connections between these descriptions to the algebraic descriptions.



To add a layer of rigor to this activity, consider having students switch the preimage and image and write a new sequence of transformations.

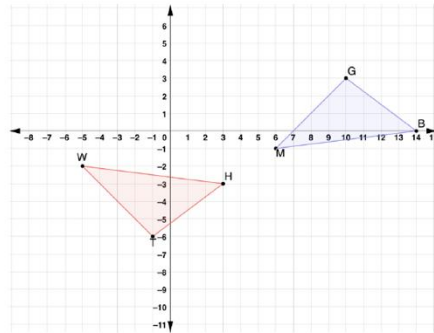
4.7.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door on describing a sequence of transformations to justify congruence. This activity should be done independently to collect student-level performance data.



4.7.5 Wrap-Up: Ticket Out the Door

- Triangles TWH and GBM are shown on the coordinate grid.



Notice this question intentionally omits the identification of preimage and image, as either triangle could be the preimage for the congruence statement to still be true.

- Use written descriptions or algebraic descriptions to describe a sequence of two transformations that will map the two triangles onto each other.

To map $\triangle GBM$ onto $\triangle TWH$, then perform a reflection across the x -axis, $(x, y) \rightarrow (x, -y)$ and a horizontal translation left 11 units and vertical translation down 3 units, $(x, y) \rightarrow (x - 11, y - 3)$.
To map $\triangle TWH$ onto $\triangle GBM$, then perform a reflection across the x -axis, $(x, y) \rightarrow (x, -y)$ and a horizontal translation right 11 units and vertical translation down 3 units $(x, y) \rightarrow (x + 11, y - 3)$.



Consider prompting students who finish early to determine if multiple sequences could map the same images or how changing which triangle is the preimage would change the sequence of transformations.

- Complete the table that shows which vertices were mapped to each other.

$\triangle TWH$	$\triangle GBM$
T	G
W	M
H	B

- Complete the congruence statement.
 $\triangle TWH \cong \triangle GMB$